C PROGRAMMING — ASSIGNMENT 1

Task 3: Define the Constants

Marks: 3 | Total assignment: 50 marks

Explanation

Named constants centralize the tariff so a revision is made in one place. #define is processed by the preprocessor and does not declare a typed object; const variables have a type, obey scope rules, and are easier to inspect in a debugger. enum is appropriate for integral named choices such as customer categories or status codes, not for monetary values that may exceed int. Use uppercase names with underscores for macros.

Tariff calculation

The fixed charge is 5000 paise. The first 100 units are free; units 101–200 cost 225 paise each; units above 200 cost 450 paise each. For 150 units, the bill is 5000 + 50×225 = 16250 paise = Rs 162.50. Changing RATE2_PAISE from 225 to 230 changes all charges for the second slab consistently.

C Source Code / Implementation

```
#include <stdio.h>
#define SLAB1_UNITS 100
#define SLAB2_UNITS 200
#define RATE1_PAISE 0
#define RATE2_PAISE 225
#define RATE3_PAISE 450
#define FIXED_CHARGE_PAISE 5000L

long compute_bill(int units) {
    long paise = FIXED_CHARGE_PAISE;
    if (units > SLAB2_UNITS)
        paise += (long)(units - SLAB2_UNITS) * RATE3_PAISE;
    if (units > SLAB1_UNITS) {
        int second = (units < SLAB2_UNITS ? units : SLAB2_UNITS) - SLAB1_UNITS;
        paise += (long)second * RATE2_PAISE;
    }
    /* Units in the first slab cost RATE1_PAISE (zero here). */
    return paise;
}
int main(void) {
    int tests[] = {0, 100, 150, 200, 350};
    size_t n = sizeof tests / sizeof tests[0];
    for (size_t i = 0; i < n; ++i) {
        long p = compute_bill(tests[i]);
        printf("%3d units -> Rs %ld.%02ld\n", tests[i], p / 100, p % 100);
    }
    return 0;
}
```

Submission note

Compile and run the code on your own system, save actual terminal output/screenshots, and include them in your report. Review the code and be prepared to explain it during the viva. Any machine-specific measurements must come from your own execution.